



Applied Data Mining

**— Week 10—
Spring 2026**

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Outline

- Cluster Analysis: Basic Concepts 
- Partitioning Methods
- Hierarchical Methods
- Summary

What is Cluster Analysis?

- **What is clustering?**

- A process of partitioning a set of data objects (or observations) into subsets. Each subset is a cluster.
- Cluster analysis and clustering are usually used interchangeably.

- **Difference between Cluster analysis and clustering**

- Sometimes CA can be used more broadly to encompass the entire process of exploring and understanding data through grouping, while "clustering" might refer to the specific algorithmic or technique implementation of grouping.

CA = complete analysis process

Clustering = implementation method

- **Cluster: A collection of data objects**

- Similar (or related) to one another within the same group
- Dissimilar (or unrelated) to the objects in other groups

What is Cluster Analysis?

- Dissimilarities and similarities are assessed based on the **attribute values** describing the objects and often involve distance measures.
- **Unsupervised learning**: no predefined labels or classes (i.e., *the system learn patterns automatically from the data/observation.*)
- In clustering: No labels exist, Algorithm itself finds groups
- **Typical applications**
 - As a **stand-alone tool** to get insight into data distribution
 - As a **preprocessing step** for other algorithms

Clustering for Data Understanding and Applications

- Information retrieval: document clustering
- Land use: Identification of areas of similar land use in an earth observation database
- Marketing: Help marketers to discover distinct groups among their customers, and then use this knowledge to develop targeted marketing programs
- City-planning: Identifying groups of houses according to their house type, value, and geographical location

Quality: What Is Good Clustering?

- A good clustering method will produce high quality clusters
 - High intra-class similarity: **cohesive** within clusters
 - Low inter-class similarity: **distinctive** between clusters
- The quality of a clustering method depends on
 - The similarity measure used by the clustering method
 - Its implementation
 - Its ability to discover some or all of the hidden patterns

Measure the Quality of Clustering

- Dissimilarity/Similarity metric
 - Similarity is expressed in terms of a distance function/calculation, typically metric: $d(i, j)$

$$\rho(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

This is called the Manhattan Distance or City Block Distance.

- Smaller distance → more similarity
- Larger distance → less similarity
- The definitions of distance functions are usually rather different for interval-scaled, boolean, categorical, ordinal ratio, and vector variables
- Quality of clustering:
 - There is usually a separate “quality” function that measures the “goodness” of a cluster.

Major Clustering Approaches

- Partitioning approach:
 - Construct various partitions and then evaluate them by some criterion, e.g., minimizing the sum of square errors
 - Typical methods: k-means, k-medoids, CLARANS
- Hierarchical approach:
 - Create a hierarchical decomposition of the set of data (or objects) using some criterion
 - Typical methods: Diana, Agnes, BIRCH, CAMELEON
- Density-based approach:
 - Based on connectivity and density functions
 - Typical methods: DBSACN, OPTICS, DenClue
- Grid-based approach:
 - based on a multiple-level granularity structure
 - Typical methods: STING, WaveCluster, CLIQUE

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Partitioning Algorithms: Basic Concept

- This method organizes the objects of a set into several exclusive groups or clusters.
- We can assume that the number of clusters is given as background knowledge.
- This parameter is the starting point for partitioning methods.
- **Formally:** Given a data set D , of n objects, and k (the number of clusters to form), a partitioning algorithm organizes the objects into k partitions ($k \leq n$), where each *partition* represents a cluster.

Partitioning Algorithms: Basic Concept

- Given k , find a partition of k clusters that optimizes the chosen partitioning criterion
 - Global optimal: exhaustively enumerate all partitions
 - Check every possible grouping
 - Select the best one.
 - Heuristic methods: *k-means* and *k-medoids* algorithms
 - *k-means* (MacQueen'67, Lloyd'57/'82): Each cluster is represented by the center of the cluster
 - *k-medoids* or PAM (Partition around medoids) (Kaufman & Rousseeuw'87): Each cluster is represented by one of the objects in the cluster

The K-Means Clustering Method



Given k , k -means can be implemented in four steps:

1. Randomly choose k centroids (or points that will be the center of your clusters) in d -space. Try to make them near the data but different from one another.
2. Then assign each data point to the nearest centroid or find the closest cluster center for each item, and assign it to that cluster.
3. Recalculate the cluster centroid as the mean of items/points, for the newly-assigned items in the cluster.

The K-Means Clustering Method

- 4. **Repeat the preceding two steps until**
 - For a fixed number of iterations
 - The assignments don't change
 - The change is very little.



Any Question